

Preprocessing
resize 128×128,
color-space convert

125-d color-statistical
features (RGB+HSV+Lab,
histograms, ripeness idx.)

Logistic
Regression

Random
Forest

Extra
Trees

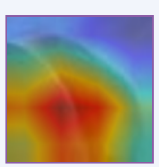
K-Nearest
Neighbors

HHO ensemble-
weight optimizer
(Eq. 1-3)

Tomato
Quality
Index +
grade
(Eq. 4)

ResNet18 CNN
(fine-tuned,
ImageNet init.)

Grad-CAM
visual
explanation



4-class grade
+ heatmap
overlay